Phase 1: Proposal

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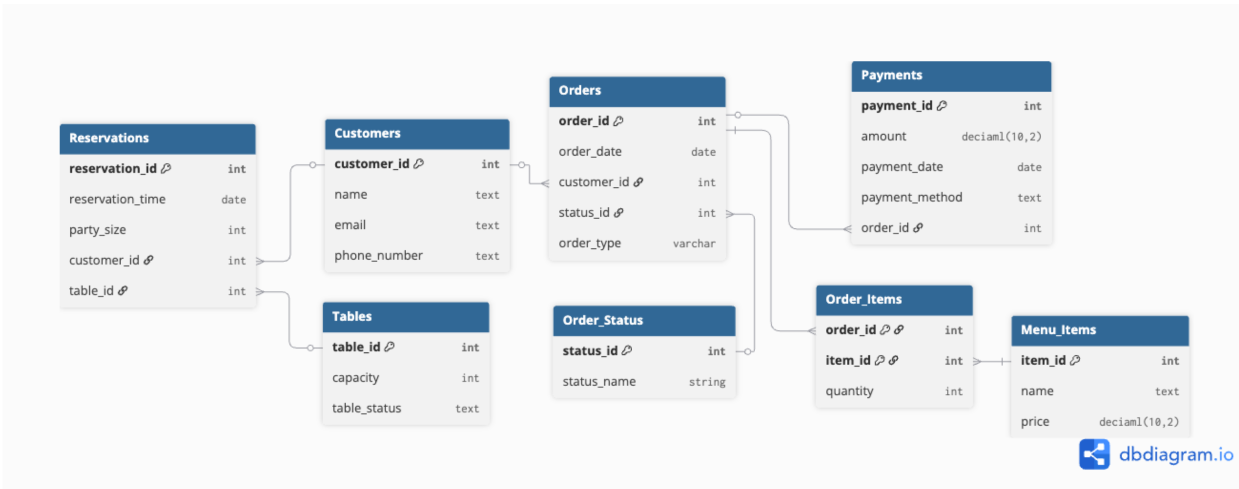
System Description:

This database will be an order management system that will support the daily operations of a restaurant. The primary users of this system will be restaurant staff such as cashiers, servers, and managers, as well as customers through online ordering or table reservations. This database will track customer data, orders, menu items, payments, and reservations centralizing them into one relational database. This allows for an effective restaurant workflow by reducing manual errors, providing a clear overview of all operational activities, and improving coordination between different roles within the restaurant. Additionally, this database provides a clear progression of orders from creation to completion, allowing accurate status tracking.

Scope:

- The system will store customer information
- The system will allow to create orders for online, walk-in, and phone.
- The system will record menu items included in each order.
- The system will track the status of each order.
- The system will process and store payment information for each order.
- The system will manage table reservations and assign tables to customers.

UML Overview:



Entities:

- Customers (customer_id PK, name, email, phone_number)
- Order_Status (status_id PK, status_name)
- Tables (table_id PK, capacity, table_status)
- Orders (order_id PK, order_date, customer_id FK, status_id FK, order_type)
- Menu_Items (item_id PK, name, price)
- Payments (payment_id PK, amount, payment_date, payment_method, order_id FK)
- Reservations (reservation_id PK, reservation_time, party_size, customer_id FK, table_id FK)
- Order_Items (order_id PK/FK, item_id PK/FK, quantity)

Business Rules / Constraints:

- Each customer email must be unique
- Each order must belong to one customer
- Each order must have one valid order status:

    - Received
    - Processing
    - Finished
    - Ready for pick up
    - Picked Up for online orders.
- Each order must have one valid order type:
    - Online
    - Walk-in
    - Phone Order.
- Each order must have at least one item in the menu
- The quantity of each item in an order must be greater than 0
- The price of each menu item must be greater than 0
- Each payment must be linked to exactly one order
- A payment amount must be greater than 0
- A table cannot be double-booked in the same time frame
- Each reservation must be linked to exactly one customer and one table
- The party size of a reservation must not exceed the table capacity